

A Research and Career Plan for Doctoral Study in Sustainable Tropical Building Science

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Research Focus: Energy-Efficient Passive Envelopes for Tropical Mid-Rise Buildings

This plan outlines the research I intend to carry out during my doctoral study and the academic path I aim to follow afterwards. It builds on my master's research, which identified a fundamental aerodynamic limit in integrated passive envelopes for humid tropical climates, and sets out how the doctoral work will explain that limit, resolve it, and translate the outcome into lasting academic and societal contribution.

1. RESEARCH OBJECTIVE

My doctoral research aims to explain the aerodynamic mechanism that limits integrated passive envelopes in humid tropical climates, and to develop and validate a re-engineered, decoupled envelope framework that resolves the conflict between sensible cooling and latent (humidity) load for tropical mid-rise offices. The work advances from detecting this limit – established through my master's research – to explaining it through high-resolution computational fluid dynamics (CFD), and ultimately to resolving it through optimization and validation. In doing so it directly serves the wider goal of reducing the energy use and carbon emission of rapidly expanding tropical office building stock.

2. RESEARCH PLAN & PUBLICATION TARGETS

The research is structured in three connected phases, each addressing one research question and targeting one peer-reviewed publication. The progression is cumulative: mechanism → resolution → validated framework.

PHASE 1 · MECHANISM

Explain why integrated passive envelopes short-circuit aerodynamically

Using high-resolution CFD, I will resolve the cavity and tower flow field that network-averaged airflow modelling cannot capture, explaining the flow-physics mechanism by which the heated façade monopolizes airflow and suppress fresh-air delivery. This phase establishes and validates the computational model that underpins the entire project → **TARGET OUTPUT: 1 PEER-REVIEWED Q1**

(SCIE PAPER).

PHASE 2 • RESOLUTION & OPTIMIZATION

Re-engineer the envelope geometry to break the short-circuit

Focusing on the aerodynamically active variables, I will use CFD-based multi-objective optimization – accelerated by machine-learning surrogate models – to identify geometric configurations that restore genuine fresh-air delivery while preserving solar shading and thermal buffering. This phase delivers the optimized configuration at the core of the proposed framework → **TARGET OUTPUT: 1 PEER-REVIEWED Q1 (SCIE PAPER).**

PHASE 3 – VALIDATED DECOUPLED FRAMEWORK

Implement and validate the decoupled framework

I will integrate the optimized passive configuration with targeted active mechanical supplementation and evaluate, through coupled simulation, the extent to which the decoupled framework resolves the sensible-latent conflict – its first implementation and test. Where scope permits, a bounded generalization study across representative tropical urban climate (e.g. Kuala Lumpur and Singapore) will test the transferability of the framework and its diagnostic metric → **TARGET OUTPUT: 1 PEER-REVIEWED Q1 (SCIE PAPER).**

3. HOW I WILL REALIZE THE RESEARCH

To deliver this plan, I will build on the methodological foundation established in my master's work (building-performance simulation and airflow analysis) and extend it with the advanced computational skills the doctoral work requires: high-resolution CFD, multi-objective optimization, and machine-learning surrogate modelling, with scripting in python. The first year is dedicated to developing CFD proficiency and constructing a rigorously verified and validated model, ensuring that every subsequent result rests on a trustworthy computational basis. Throughout, I will findings progressively through peer-reviewed publications and conference presentations rather than deferring all output to the end, building a clear and cumulative research record.

4. INDICATIVE RESEARCH TIMELINE

Year	Focus	Output
Year 1	Literature synthesis; CFD competency; model build, verification & validation	Validated CFD model; phase 1 paper draft
Year 2	Core CFD simulations; mechanism analysis; surrogate-model development	Phase 1 publication; optimization pipeline

Year 3	Multi-objective optimization; optimized configuration; begin generalization	Phase 2 publication
Year 4	Framework integration & validation; generalization; synthesis & dissertation	Phase 3 publication; doctoral thesis

5. CAREER GOALS AFTER THE DOCTORATE

My long-term aspiration is an academic career contributing to Indonesia's capacity in sustainable tropical building design.

ON COMPLETION OF THE Ph.D.

Enter an academic teaching and research role in Indonesia

I intend to take up a faculty position, contributing to architectural and building-science education while establishing an independent research agenda in tropical building performance – carrying the doctoral work back to the building stock and climate it was designed to serve.

SUBSEQUENTLY

Strengthen research capacity through post-doctoral work

Once established in an academic role, I intend to pursue post-doctoral research – internationally (e.g. previous academic institutions at which I obtained either a Master's or Doctoral degree) to deepen my research independence, publication record, and collaborative networks, as a deliberate capacity-building step.

LONG-TERM CONTRIBUTION

Build a research program on tropical building decarbonization

Over the longer term, I aim to lead research and contribute to evidence-based building standards for energy-efficient, low-carbon design in tropical climates – translating doctoral and post-doctoral work into lasting academic and policy impact in Indonesia.

Timeline is indicative; phase duration may adjust to program structure and the scope ultimately agreed with my doctoral supervisor.